

Survival Model Evaluation: `chicago_licences`

Fitted with the strategy-survival-model pipeline; methodology validated separately against synthetic ground truth

Source data	<code>datasets/chicago_licences.csv.gz</code>
Rows	239,721 (82.2% with the ending observed, 17.8% censored)
Start dates	2002-01-02 to 2026-07-31
Evaluation	5 expanding-window temporal folds
Source of figures	<code>reports/chicago_demo/metrics.json</code> , regenerated by the command in section 7

1. Summary

This report evaluates a survival model fitted to `chicago_licences.csv.gz`, 239,721 rows, each observed from its start date for 'licensed_days' days with 'closed' marking whether the ending was seen (82.2%) or the row was still running when observation stopped (17.8%). Median observed duration is 904 days. The model predicts, from the 289 features the deployed encoding builds out of the start-date columns (each evaluation fold refits a smaller vocabulary of its own), how long each row survives.

The like-for-like comparison between models is the fold mean: each of the 5 folds fits both models on its own training window and the fold scores average. On concordance, the share of pairs a ranking puts in the right order where 0.500 is a coin flip, the XGBoost AFT model scores 0.585 and a Cox proportional hazards baseline on the same features 0.592. The simpler baseline wins, and scoring new rows defaults to it, because the saved bundle recommends whichever model scored higher on this comparison.

Pooled over 143,833 out-of-time test rows, the AFT model scores 0.573 (95% bootstrap interval 0.571 to 0.575). The pooled figure exists for the decomposition below, which needs every test row scored on one axis, rather than as a headline. Pooling concatenates predictions from 5 separately fitted models, and although AFT medians share the time scale, each fold's model carries its own level, so cross-fold pairs blur and the pooled figure reads conservative next to the fold means. The Cox baseline's fold-relative risks cannot be pooled at all, which is why the model comparison above uses fold means for both. Most of the pooled figure reflects a row's `license_description` group rather than ranking within it. Ranking every row by its group's mean prediction alone scores 0.613, and comparisons within a group score 0.510; the results section gives the decomposition. At the 1095-day horizon its censoring-weighted Brier score is 0.239, beating the 0.250 of a no-skill forecast

that assigns every row the population average, but at the 365-day horizon it loses to that reference; section 6 covers the loss.

That bootstrap interval is narrow, and it is worth saying what it does and does not cover. It holds each fold's fitted model fixed and resamples the test rows, so it measures how precisely this run's score is pinned down by the number of rows scored, nothing more. It says nothing about how much the score would move on a different stretch of history, and the per-fold spread of 0.554 to 0.610 in Table 2 is the better guide to that. The rows are also unlikely to be independent, since rows that share a category or a stretch of time tend to fail together, which makes the true interval wider than the one printed.

This dataset has no known generating process. Unlike the synthetic validation report, there is no oracle ceiling to say how much signal remains unclaimed, and feature attributions cannot be checked against a true mechanism. What carries over from the synthetic validation is the pipeline itself: the same temporal folds, label re-censoring, likelihood-based selection, and calibration checks, verified there against known answers.

2. Data

The dataset is `datasets/chicago_licences.csv.gz` : 239,721 rows, each observed from its start date. Start dates run 2002-01-02 to 2026-07-31. 82.2% of rows have their ending observed and 17.8% are censored, meaning the row was still open when observation stopped, so its full duration is unknown; median observed duration is 904 days. Text columns are one-hot encoded automatically. These columns hold numeric codes rather than quantities and were forced to categorical as well: `ward`, `community_area`, `police_district`, `zip_code`.

The pipeline treats every row as observed from its own start date. It has no support for left truncation, rows whose life began before the source window opened and that would enter the data mid-life, so any such rows have to be excluded when the dataset is prepared. Whether they were, and how many, is a property of the preparation step recorded with the dataset.

For this dataset the preparation step did perform that exclusion. Licences already running when the portal's history begins enter at their first recorded renewal rather than their true start; 79,690 of them, 23 percent of the pull, were removed by keeping only licences whose earliest transaction is a genuine issue. The tell was a spike of 61,351 supposed starts dated 2002, against roughly 14,000 genuine first issues in a normal year. The rule and the counts are recorded in `scripts/run_prepare_chicago.py` . Licence types that are event-scoped by construction, the Special Event, Pop-Up, and Itinerant variants, are excluded as well; that removed 23,042 licences across 12 types, permits that typically expire within their first week. Those lives are short because the

licence says so, and a lifespan that is intent rather than failure is the wrong thing to ask a survival model about.

Figure 1 shows Kaplan-Meier survival curves by `license_description`, the coarsest structure in the outcome before any model is fitted. A Kaplan-Meier curve estimates the fraction of rows still surviving at each age, with censored rows counted for as long as they were observed.

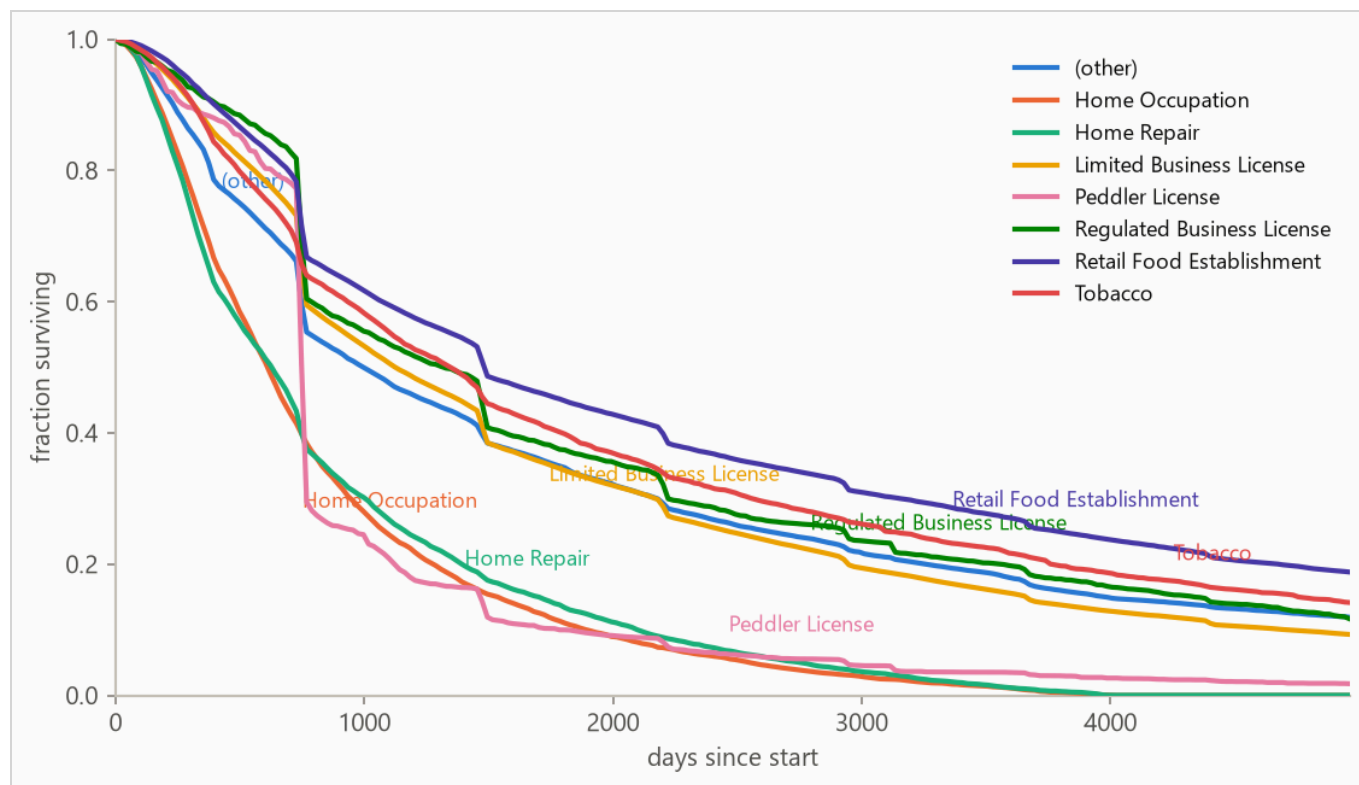


Figure 1. Kaplan-Meier survival curves by `license_description`. Groups beyond the seven most frequent are collapsed into (other) for this plot only. The vertical drop in the (other) curve at about two years, and the smaller steps at term multiples in other curves, are licence terms expiring. A business that does not renew exits at a term boundary, so endings cluster there.

3. Method

3.1 Model class

The target is a duration with incomplete observations, so the model is an accelerated failure time (AFT) model, implemented as XGBoost with the `survival:aft` objective. Censoring enters through interval labels, where an observed death is the interval $[t, t]$ and a censored row is $[t, \text{infinity})$.

Regression on lifetime was rejected because censored rows have no target, and dropping them discards the longest-lived rows, which biases the model toward pessimism. Classification at a fixed horizon handles censoring awkwardly and answers only one question. AFT keeps every row and returns a full time distribution, which collapses to any horizon of interest.

Concretely, the fitted model predicts one number per row, the median survival time in days. The full curve comes from wrapping a log-normal distribution of fitted width around that median, so the probability of surviving any horizon is a point read off that curve. The width is a single value shared by every row; how it is selected and calibrated is covered in section 3.3.

Numeric features pass through as-is, missing values included, which XGBoost handles natively; the Cox baseline receives train-window median imputation. Text columns are one-hot encoded, and the encoding vocabulary (which levels exist, which rare levels collapse) is refit on each fold's training window, so a fold's features reflect only what was on file by its split date. The deployed model's recipe is fit on the full window, whose past is legitimately the whole file.

3.2 Temporal validation and label re-censoring

Rows are ordered by start date and evaluated with 5 expanding-window folds. The earliest 40% of rows is burn-in that is only ever trained on. Every fold trains only on rows that started before its split date and tests on the next block, so training sets grow from 95,870 to 210,947 rows.

Training labels are re-censored at each split date. A row started two years before a split may have died three months after it, and its recorded label contains that future death. Any model trained at that split date could only have known the row was still running, so every post-split death is rewritten as a censoring at the split. Omitting this step raises measured scores while silently importing future information, which makes it the most consequential detail in the pipeline. It is implemented as a standalone function (`cv.recensor`) with dedicated tests, because it transfers unchanged to any duration problem on operational data.

Test labels use the full observation window, because the restriction exists to keep the future away from the model, not away from the scorer.

3.3 Hyperparameter selection and calibration

Hyperparameters are selected once, on the first fold's training window, using an inner temporal split and held-out censored log-likelihood. Likelihood rather than concordance drives selection, and the reason is a failure this pipeline recorded rather than avoided.

An earlier version selected on concordance and looked correct, with ranking metrics landing where they ultimately did. It then lost to a no-skill reference forecast on 365-day Brier score, the mean squared error of a probability forecast. A concordance index is invariant to the predictive scale, so the search had no reason to

prefer a usable distribution width and settled on one far too wide. Nothing in the training loop objected, because nothing in it measured the width. Selection now uses likelihood, which penalizes a broken scale, and the predictive log-normal scale is calibrated on a temporal tail slice by a probe model that never trained on those rows; the final model is then refit on the full window, including that slice, and carries the probe-calibrated scale. The two scales do different jobs. The loss scale is a setting inside the training objective, while the predictive scale is the width of the probability curves the model actually reports, measured afterward on rows the fit never used. The selected loss scale was 1.2, and the calibrated predictive scale came out at 0.88.

The general lesson holds beyond this project. A model can rank well and misstate probabilities at the same time, and the only defense is evaluating both.

3.4 Baselines

A Cox proportional hazards model fitted on the same features is the standard survival alternative and answers whether the gradient-boosted model was necessary.

4. Results

4.1 Discrimination

Pooled over 143,833 out-of-fold rows, with 95% percentile bootstrap intervals over 500 resamples:

Table 1. Concordance index by method (Harrell's C is its standard estimator). Higher is better; 0.500 is a coin flip. Cox appears as a fold mean only, because its per-fold risk scores share no scale to pool.

METHOD	CONCORDANCE (HARRELL'S C)	95% INTERVAL
XGBoost AFT (pooled)	0.573	0.571 to 0.575
XGBoost AFT (fold mean)	0.585	not resampled
Cox proportional hazards (fold mean)	0.592	not resampled

On this dataset the Cox baseline scores 0.5918 against the boosted model's 0.5849 on fold-mean concordance, a margin that sets the default for scoring new rows without proving a real difference. Both fitted models are saved and the Cox model is the one this run recommends. Reporting the near-tie, rather than presenting the boosted model as the result, is the point of carrying a baseline at all. Each fold refits the Cox model, and its risk scores are relative ones centred on that fold's own training window, so they are meaningful within a fold but carry no common scale across folds; the fold-mean rows are the like-for-like comparison and the pooled figure is not comparable to them.

The per-fold spread deserves as much attention as the mean. The weakest window is fold 3, starting 2013-11-27, at 0.554. That window's weakness has a measurable cause. Its test block sits across a shift in the city's category mix. The Home Occupation and Home Repair categories, carrying 11 percent of its training rows, stop appearing in the test block entirely, Regulated Business License nearly triples its share, and 2 percent of test rows are in categories the training window never saw. Re-selecting hyperparameters on this fold's own training window closes about a third of the gap to the Cox baseline, and the remainder is consistent with the boosted model leaning harder than the linear baseline on category structure that had just moved. Any use of the pooled figure should carry that range: there are stretches of history where the model ranks these rows substantially worse than its headline number suggests.

The pooled concordance also decomposes by `license_description`. Ranking every row by its group's mean prediction alone scores 0.613, and comparisons restricted to rows in the same group score 0.510, pair-weighted across 65 groups with at least 50 rows and 10 observed endings. The distance of the within-group figure above 0.500 is the row-level skill; the rest of the pooled figure is group membership. Both numbers come from the same out-of-fold predictions as the pooled figure. Stated plainly, the group means alone outscore the full model's pooled figure; the within-group variation adds ranking skill inside a group and costs a little ordering across groups.

Table 2. Per-fold results. Censoring is the share of each fold's test rows whose ending was not observed.

FOLD	SPLIT DATE	TRAIN N	TEST N	CENSORED	AFT	COX
1	2009-05-18	95,870	28,767	7%	0.600	0.595
2	2012-03-27	124,601	28,767	14%	0.558	0.558
3	2013-11-27	153,398	28,767	18%	0.554	0.599
4	2017-08-07	182,164	28,766	29%	0.601	0.592
5	2021-11-02	210,947	28,766	67%	0.610	0.615

Figure 2 plots the per-fold concordance from Table 2; each fold's censoring mix is in the table.

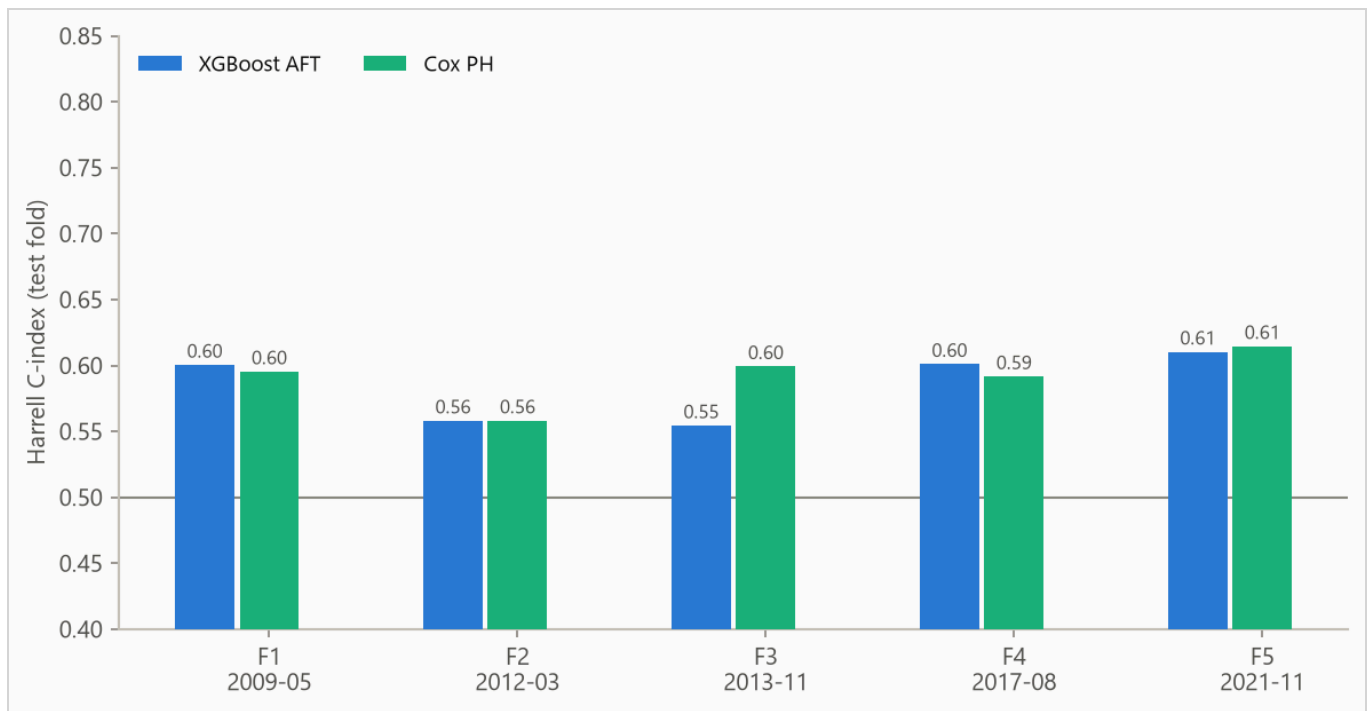


Figure 2. Concordance by fold, from 0.554 to 0.610. Folds differ in censoring mix, so cross-fold comparisons are indicative rather than exact.

4.2 Calibration

Predicted median survival times are converted to survival probabilities under the calibrated log-normal, then checked two ways. The Brier score is the mean squared error of a probability forecast, lower being better, and here it is weighted by the inverse probability of censoring (IPCW), so that rows whose outcome was censored away do not bias the result. The no-skill reference assigns every row the pooled Kaplan-Meier marginal, ignoring all features.

Table 3. IPCW Brier score by horizon. Lower is better.

HORIZON	AFT	COX	NO-SKILL MARGINAL
365 days	0.097	0.088	0.081
1095 days	0.239	0.231	0.250
1825 days	0.217	0.212	0.227

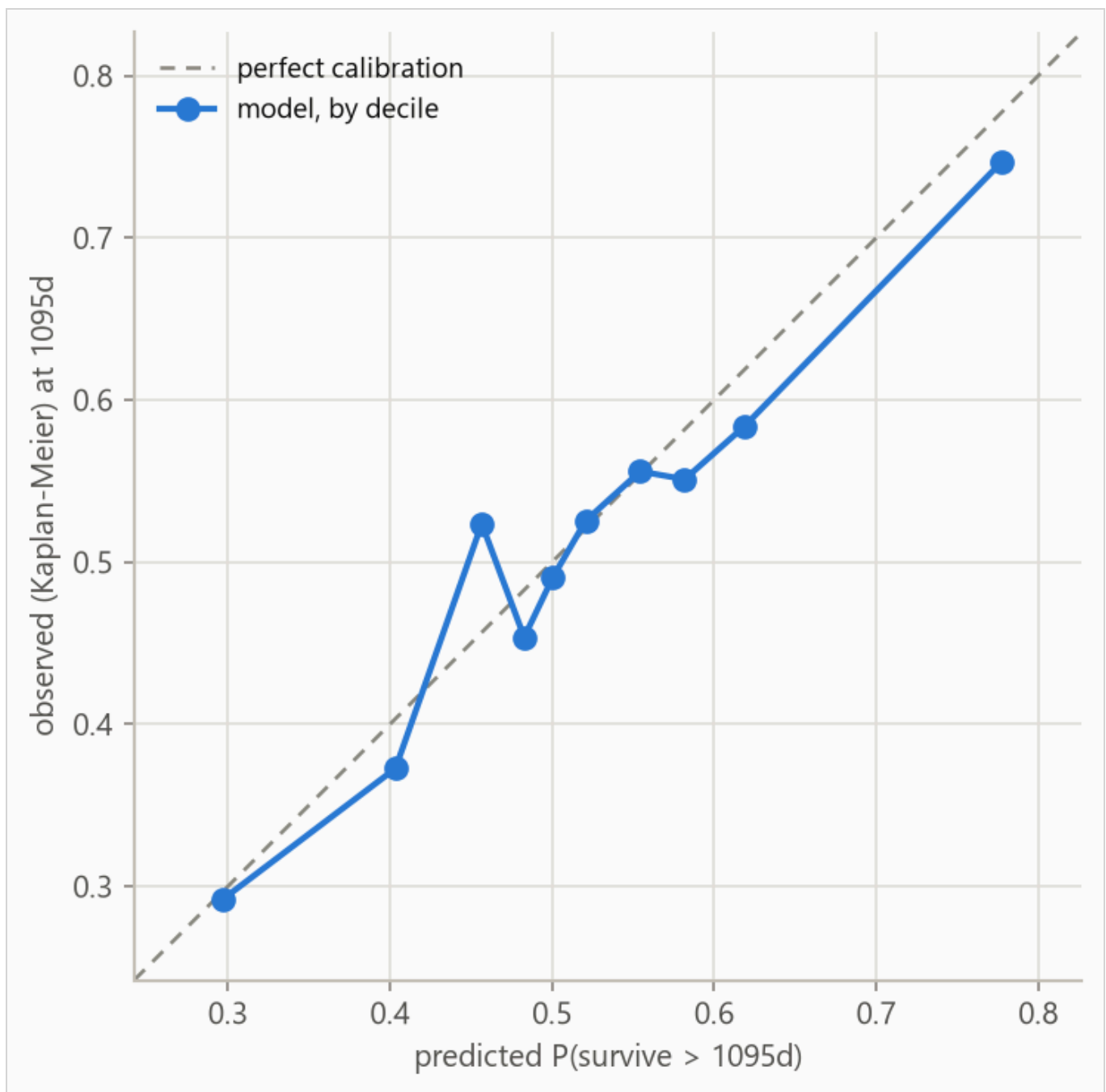


Figure 3. Predicted against observed survival at 1095 days, by predicted decile. Observed frequencies are Kaplan-Meier estimates within each bin, so censored rows contribute correctly. The largest deviation is 0.067 in decile 3, and it is a real miss, not display noise.

Table 4. Decile calibration at 1095 days, the values plotted in Figure 3.

DECILE	N	PREDICTED	OBSERVED (KM)	DEVIATION
1	14492	0.297	0.292	-0.005
2	14933	0.403	0.373	-0.031
3	13805	0.457	0.524	+0.067
4	15540	0.483	0.454	-0.029
5	13533	0.501	0.491	-0.010
6	14076	0.521	0.526	+0.005
7	14318	0.554	0.556	+0.002
8	15061	0.582	0.551	-0.031
9	13874	0.619	0.584	-0.035
10	14201	0.777	0.747	-0.031

One caveat on the decile table. The observed value inside each decile is a Kaplan-Meier estimate, and when a decile's last observed row falls short of the 1095-day horizon the estimate carries its last value forward. In small or heavily censored deciles the observed column is softer than its three decimals look, and deviations there should be read accordingly.

5. What the model uses

Feature attributions (SHAP values, for SHapley Additive exPlanations) are computed on the log scale of survival time, so a value of +0.3 multiplies predicted survival time by about 1.35 and negative values shorten it.

On this data the attributions are descriptive only. There is no known mechanism to check them against, and correlated features can split credit in ways that reflect the model's internal choices as much as the data, so read the ranking as "what this model leaned on", not as importance in the world. They are also computed on the final model's own training rows, an in-sample reading rather than out-of-fold evidence.

Figure 4 ranks features by mean absolute attribution, Table 5 lists the top eight, Figure 5 shows the per-row spread, and Figure 6 traces attribution against feature value.

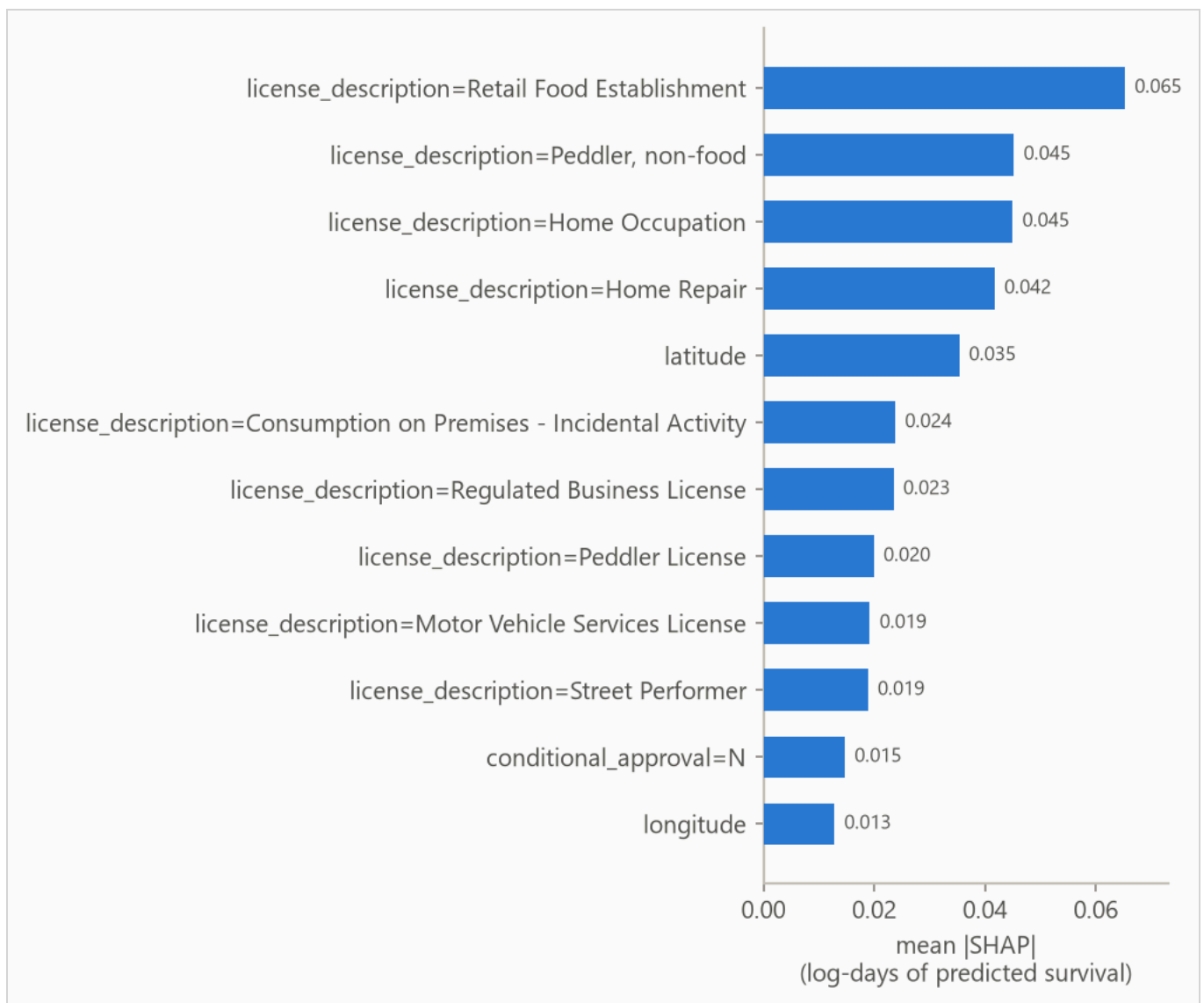


Figure 4. Mean absolute attribution by feature.

Table 5. Top eight features by mean absolute attribution.

FEATURE	MEAN ATTRIBUTION
license_description=Retail Food Establishment	0.065
license_description=Peddler, non-food	0.045
license_description=Home Occupation	0.045
license_description=Home Repair	0.042
latitude	0.035
license_description=Consumption on Premises - Incidental Activity	0.024
license_description=Regulated Business License	0.023
license_description=Peddler License	0.020

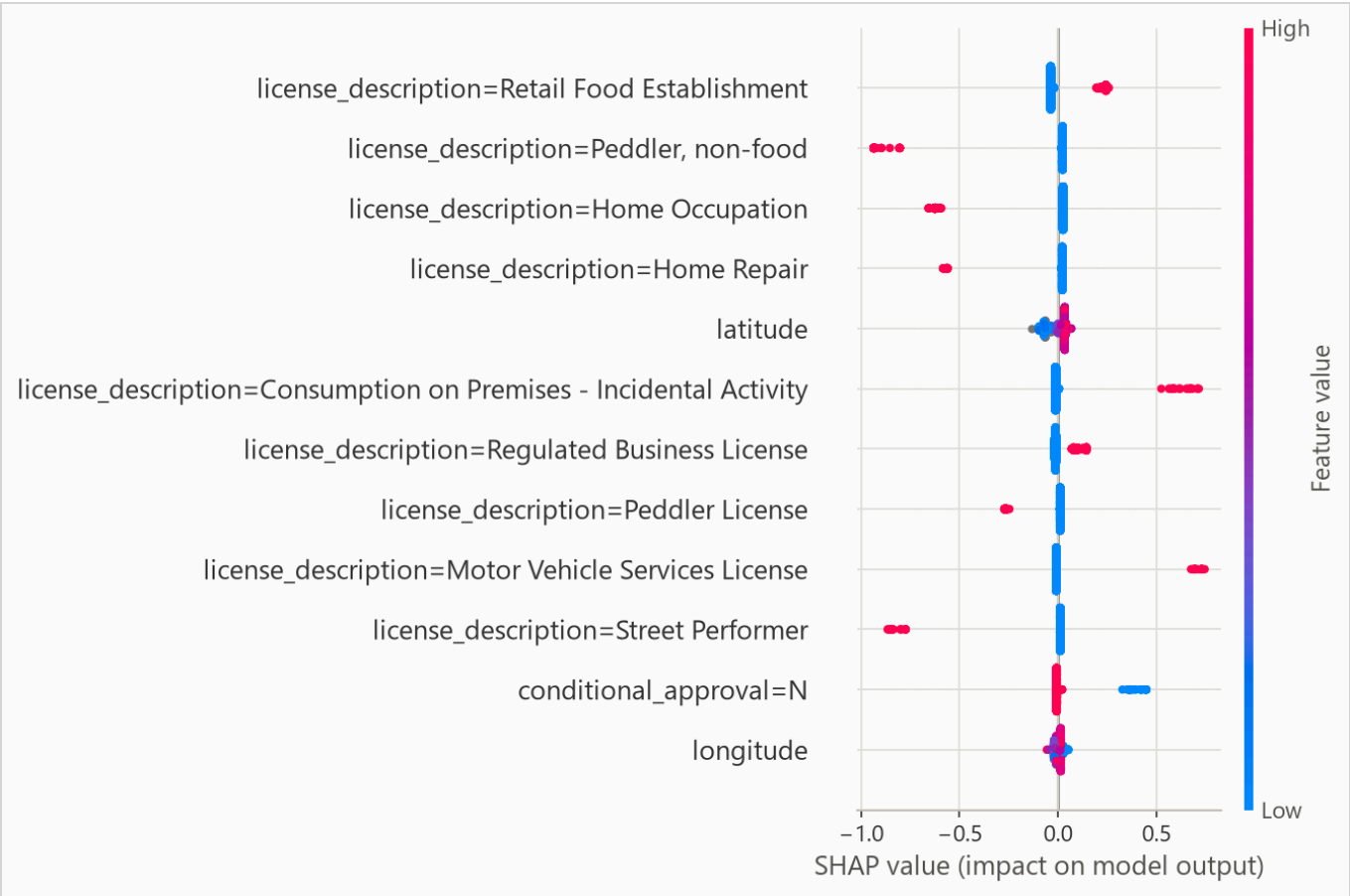


Figure 5. Per-row attributions across the explanation sample. For a yes/no feature, such as a one-hot category flag, the high (red) end simply means the row is in that category.

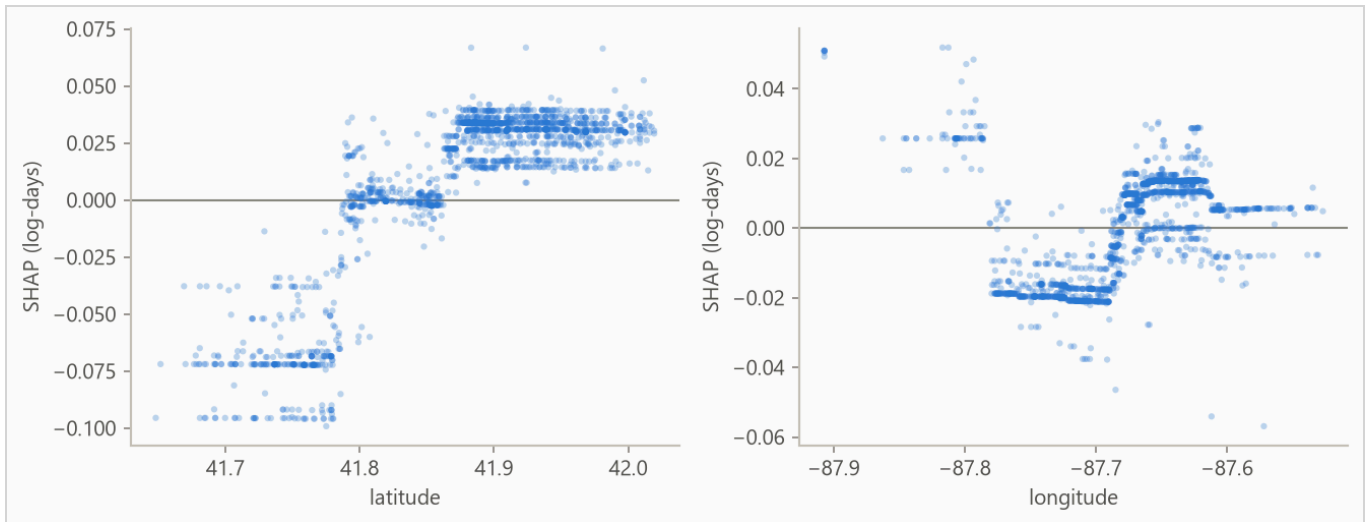


Figure 6. Attribution against feature value for the strongest numeric features; category flags carry no shape and stay in the ranking above. A run short on numeric features fills the grid with flags instead, two columns of dots per panel.

6. Limitations

No ground truth. Nothing bounds how much predictable signal this dataset contains, so the concordance above cannot be placed relative to a ceiling the way the synthetic validation could.

Absolute probabilities are not usable at 365 days. At those horizons the model's censoring-weighted Brier score does not beat a no-skill forecast. Discrimination and calibration are separate qualities. The ranking carries signal while the survival probabilities do not, so use this model to order rows, not to act on absolute probabilities at those horizons. The usual cause under temporal evaluation is training windows whose re-censored follow-up is far shorter than the horizon, which forces the fitted log-normal to extrapolate beyond anything it saw.

Censoring may be informative. The evaluation assumes rows stop being observed for reasons unrelated to their risk. If observation ended early on rows that were about to end anyway, survival estimates are biased upward. Whether that holds here depends on how this dataset was collected.

Endings near the cutoff are provisional. Where no cancellation exists, a licence's end is its latest recorded expiry, so a licence whose term ran out shortly before the cutoff but that renews late is recorded as an observed closure. Deaths dated close to the cutoff therefore include some licences that will turn out to have survived; how many is not measured here.

Every row shares one curve shape. The predictive scale is a single fitted number, so the model shifts each row's survival curve earlier or later but never widens or narrows it, and two rows with the same predicted

median receive identical probabilities at every horizon. Letting the width vary by row would need a model that predicts it, which is future work.

Harrell's concordance is biased under heavy censoring. The final fold is 67% censored, where an inverse-probability weighted concordance would be preferable. Fold-to-fold comparisons in Table 2 should be read with that in mind.

7. Reproducing this run

```
python scripts/run_fit_evaluate.py --data datasets/chicago_licences.csv.gz --name chicago_licences
python scripts/run_build_report.py --run reports/chicago_demo
```

Generated from `reports/chicago_demo/metrics.json` by `scripts/run_build_report.py --run .` 239,721 rows, 143,833 out-of-time test rows.